

Array Selection for Wearable Ultrasound Dysphagia Biofeedback Monitor Device

Alessia Bruno¹, Xanthe Miller¹, Ourania Varsou², Emma Harris³, Sandy Cochran¹

¹James Watt School of Engineering, University of Glasgow, Glasgow, UK

²Anatomy Facility, School of Medicine, Dentistry and Nursing, University of Glasgow, UK

³Imaging for Radiotherapy Adaptation, Institute of Cancer Research, Sutton, UK

Background, Motivation and Objective

Wearable ultrasound is an emerging technology in medical imaging, enabling point-of-care monitoring in real time for extended periods. In the field of speech and language therapy (SLT), a specific opportunity exists for ultrasound to monitor dysphagia (swallowing disorder) in real-time in hospital and home environments. Bulky clinical systems are presently used with intrusive attachment frames. Transitioning to a small, wearable system is attractive but requires an optimal selection of the array, in terms of flexibility and frequency range, to balance image quality and penetration depth for dysphagia biofeedback monitoring in SLT.

Statement of Contribution/Methods

This study evaluated the suitability of different ultrasound transducers. Three arrays were characterized using the Verasonics Vantage 128 research system: (i) 14 MHz rigid piezocomposite (Vermon, France); (ii) 28 MHz single-crystal (Vermon); and (iii) 25 MHz flexible thin-film (Novosound, UK) (Fig.1). Each array, without a casing, was connected to the Verasonics system via flexible PCBs and interposers. Performance was evaluated through standard characterization methods: impedance analysis, pulse-echo tests, field scans in a scanning tank, tests on phantoms and finally on fresh frozen cadavers.

Results/Discussion

Whilst many publications suggest flexible arrays have advantages for wearable, the human body itself provides sufficient compliance. Our characterization also revealed challenges using the thin-film flexible array, especially because of element dropout and limited penetration depth, which compromised image quality for deep neck structures. The 28 MHz single-crystal array had no dropout, and its penetration depth was higher but still inadequate. In contrast, the 14 MHz rigid array demonstrated itself to be more suitable, ruling out flexible and high frequency options.

Our findings suggest that, despite the flexible transducer ergonomic advantages and the resolution enhancement from a high frequency array, a high-performance rigid array, at the high end of the conventional 2 – 15 MHz frequency range, currently provides a better option for dysphagia monitoring. These results support the ongoing development of a wearable device based on the integration of an optimized rigid transducer with a bespoke case for application in SLT.

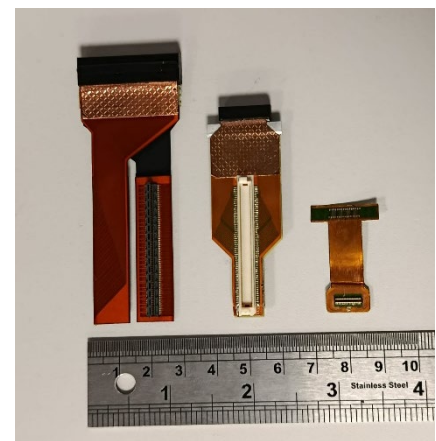


Figure 1: Evaluated arrays (left to right): 14 MHz rigid, 28 MHz single-crystal, and 25 MHz flexible with integrated PCB. Ruler for scale.